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PAPER-1 (COMPULSORY)

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GENERAL PAPER

[19 DEC 2023 (E)]

The following table shows the number of employees working in five different colleges A-E, alongwith the ratio of Males to Females [M : F] among them. Based on the data in the table, answer the questions that follow.

College-wise Details of Employees

College	Number of Employees	Ratio
		M : F
A	3040	13 : 6
B	3360	4 : 3
C	2920	9 : 11
D	2760	10 : 13
E	2840	13 : 7

- What is the total number of male employees taking all the Colleges together?
 (1) 8120 (2) 8240
 (3) 8360 (4) 8560
- What is the average number of female employees taking all the colleges together ?
 (1) 1312 (2) 1372
 (3) 1336 (4) 1292
- The number of male employees working in Colleges A and C together is approximately _____% more than the number of female employees working in colleges B and D together.
 (1) 9 (2) 15
 (3) 11 (4) 13
- What is the ratio of female employees working in College D to that in College E ?
 (1) 7 : 4 (2) 780 : 497
 (3) 390 : 491 (4) 9 : 5
- The number of employees working in College C is approximately _____% more than that in College D.
 (1) 8 (2) 6
 (3) 4 (4) 10
- An economics teacher wants to create an infographics to explain supply and demand to her students.
 Which of the following tools would be most suitable?
 (1) Edpuzzle (2) Canva
 (3) Quizlet (4) Mindmeister

7. Given below are two statements :

Statement I : Norm referencing is used to interpret a score of an individual by comparing it with those of other individuals.

Statement II : Criterion referencing is used to interpret a person's performance by comparing it to some specified behavioural domain.

In the light of the above statements, choose the most appropriate answer from the options given below :

- Both **Statement I** and **Statement II** are correct
 - Both **Statement I** and **Statement II** are incorrect
 - Statement I** is correct but **Statement II** is incorrect
 - Statement I** is incorrect but **Statement II** is correct
8. Which of the following are the elements of observational learning ?

- Production
- Attention
- Identification
- Retention
- Motivation

Choose the **correct** answer from the options given below :

- (B), (C) and (D) Only
 - (A) and (E) Only
 - (A), (B), (D) and (E) Only
 - (C) Only
9. In 'VARK' model of learning styles, 'A' stands for _____.
 (1) Auspicious (2) Augmentary
 (3) Authentic (4) Aural
10. When a very bright light is incident on our eyes, we immediately close our eyelids.
 This is an example of _____.
 (1) Biological instincts (2) Reflex actions
 (3) Desired instincts (4) Intrapersonal instincts
11. Identify the disadvantages of open ended questions in survey research.
 A. The questions do not suggest certain kinds of answer to respondents.
 B. They are time consuming for interviewers to administer.
 C. They require greater effort from respondents.
 Choose the **correct** answer from the options given below :
- (A) and (B) Only
 - (B) and (C) Only
 - (A) and (C) Only
 - (A), (B) and (C)

12. In a reference written in APA style, arrange the information pertaining to the reference in correct order.

A. DOI number B. Volume number
C. Name of author D. Title of the article

Choose the **correct** answer from the options given below :

- (1) (A), (B), (C), (D) (2) (B), (A), (C), (D)
(3) (D), (C), (B), (A) (4) (C), (D), (B), (A)

13. A research study is said to have heuristic value if :

- (1) It leads to no further research
(2) It generates more research
(3) Its results lack consistency
(4) Its results lack validity

14. An entrance examination with objective type questions is conducted. For each correct answer, 1 mark is awarded and for each incorrect answer 0.25 marks are deducted. The number of marks scored by candidates who appeared in the examination, is an example of :

- (1) nominal variable (2) discrete variable
(3) ordinal variable (4) continuous variable

15. Given below are two statements :

Statement I: Medical research usually involves pain or distress in animals.

Statement II: Psychological research is usually benign and may involve little, if any, discomfort for the animals.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both **Statement I** and **Statement II** are correct
(2) Both **Statement I** and **Statement II** are incorrect
(3) **Statement I** is correct but **Statement II** is incorrect
(4) **Statement I** is incorrect but **Statement II** is correct

16. A photograph constitutes a/an :

- (1) Protective text (2) Refined text
(3) Closed text (4) Open text

17. The Buddhist concept of communication is marked by :

- (1) Mono-causality (2) Dyadic causality
(3) Multi-causality (4) Skewed causality

18. At present, cyber communication is a :

- (1) Fictitious reality (2) Palpable reality
(3) Futuristic reality (4) Conventional reality

19. The main objectives of public policy sphere communication is to involve citizens in :

- A. Idea generation
B. Policy critique
C. Innovative policy practices
D. Ignoring oppositional views
E. Promotion of stereotypes

Choose the **correct** answer from the options given below :

- (1) (A), (B) and (C) Only (2) (B), (C) and (D) Only
(3) (C), (D) and (E) Only (4) (A), (D) and (E) Only

20. Given below are two statements :

Statement I: Western communication models examined the relationship among communicative elements on an individual basis.

Statement II: The Buddhist concept of communication emphasises the contextualisation of each of the communicative elements.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both **Statement I** and **Statement II** are true
(2) Both **Statement I** and **Statement II** are false
(3) **Statement I** is true but **Statement II** is false
(4) **Statement I** is false but **Statement II** is true

21. A man, a woman and a boy can do a piece of work in 6, 9 and 18 days, respectively. How many boys should assist one man and one woman to do the work in one day?

- (1) 13 (2) 15
(3) 11 (4) 10

22. A family consists of 6 members namely A, B, C, D, E and F. B is the brother-in-law of C's mother. C is the father of F. D is the father of A and F is the grandson of D. There are two female members in the family. Which of the following pairs represent the female members in the family?

- (1) A and D (2) D and E
(3) A and E (4) B and C

23. Find the number that can replace question mark (?) in the series given below.

1.5, 5, 10.5, 18, 27.5, 39, ?, ...

- (1) 48.5 (2) 52.5
(3) 62.5 (4) 72.5

24. A child is looking for his father. He goes 100 meter towards east and then turns to his right and goes 30 meters. From there the child again turns to his right and goes for 40 meters. Where he finds his father at his uncle's place. How far is his uncle's place from the starting point of the child?

- (1) $30\sqrt{5}$ meter (2) 40 meter
(3) $40\sqrt{5}$ meter (4) $30\sqrt{10}$ meter

25. In a certain coding language :

- (a) 1000 i.e. 10^3 is coded as 1331
(b) 125 i.e. 5^3 is coded as 216

How 729 will be coded in that language ?

- (1) 1000 (2) 1728
(3) 1331 (4) 2197

26. Which of the following statements are contradictory to each other?

- A. All plants are fungi B. No plants are fungi
C. Some plants are not fungi D. Some plants are fungi

Choose the **correct** answer from the options given below :

- (1) (B) and (D) Only (2) (A) and (B) Only
(3) (C) and (D) Only (4) (A) and (D) Only

27. If the statement - "All trucks are polluting vehicles" is given as false, then which of the following statements can be immediately inferred to be true?
- Some trucks are polluting vehicles.
 - Some trucks are not polluting vehicles.
 - No trucks are polluting vehicles.
 - No polluting vehicles are trucks.
28. Which of the following statements is equivalent to the statement - "Some philosophers are not idealists."?
- Some philosophers are not non-idealists.
 - No idealists are philosophers.
 - Some non-idealists are not philosophers.
 - Some non-idealists are not non-philosophers.
29. Which fallacy is involved in the following statement "Indian pharmaceutical companies are profitable, therefore, average Indian pharmaceutical company is profitable"?
- Fallacy of Division
 - Hasty Generalisation
 - Slippery Slope
 - Fallacy of Accident
30. Given below are two statements :
- Statement I:** The Aristotelian syllogism is only inductive and formal.
- Statement II:** Nyāya syllogism is deductive - inductive and formal-material.
- In the light of the above statements, choose the most appropriate answer from the options given below :
- Both **Statement I** and **Statement II** are true
 - Both **Statement I** and **Statement II** are false
 - Statement I** is true but **Statement II** is false
 - Statement I** is false but **Statement II** is true
31. An advantage to a business that uses Intranet capabilities is that it allows employees to share the business's information with
- Customers
 - Each other.
 - Competitors
 - Former employees
32. Consider the following MS-EXCEL screenshot that shows the product code, name of the state which produced the product and Quality of the product :
- | | A | B | C | D |
|---|---------------------|--------------|----------------|---|
| 1 | Product Code | State | Quality | |
| 2 | A1 | HP | A | |
| 3 | A2 | MP | A | |
| 4 | A3 | UP | A | |
| 5 | A4 | UP | B | |
| 6 | A5 | UP | B | |
| 7 | | | | |
- Which of the following EXCEL formulae can be typed in Cell D1 to answer the question : How many different product codes form UP are of Grade "A" quality?

- A. = COUNTIFS (B2:B6, "=UP", C2:C6, " = A")
 B. = COUNTIFS (C2:C6, "=A", B2:B6, " = UP")
 C. = COUNTIFS (B2:B6, "UP", C2:C6, "A")

Choose the **correct** answer from the options given below :

- (A) and (B) Only
 - (A) and (C) Only
 - (B) and (C) Only
 - (A), (B) and (C)
33. Given below are two statements :
- Statement I:** 'Disk defragmentation' makes files that are contiguous in a disk as fragmented.
- Statement II:** A device controller that controls a device will deal with the operating system through device driver.
- In the light of the above statements, choose the most appropriate answer from the options given below :
- Both **Statement I** and **Statement II** are true
 - Both **Statement I** and **Statement II** are false
 - Statement I** is true but **Statement II** is false
 - Statement I** is false but **Statement II** is true
34. Which of the following statements regarding Artificial Intelligence (AI) is/are correct ?
- AI systems can be used in email spam filters.
 - AI systems are capable of analyzing uncertain information.
 - An AI system is a system for capturing, storing, processing and displaying data related to positions on earth's surface.
- Choose the **correct** answer from the options given below :
- (A) Only
 - (A) and (B) Only
 - (A) and (C) Only
 - (B) and (C) Only
35. Which of the following features are available in word processing software such as MS-WORD ?
- Spell checker and Thesaurus
 - Page Orientation
 - Mail Merge
 - Slide Transitions
- Choose the **correct** answer from the options given below :
- (A), (B) and (D) Only
 - (A) and (C) Only
 - (B) and (C) Only
 - (A), (B) and (C) Only
36. Which one of the following is not caused by polluted water ?
- Cholera
 - Dysentery
 - Typhoid
 - Tuberculosis
37. Given below are two statements :
- Statement I:** Toxic chemicals often injure human organs and organ systems.
- Statement II:** Toxicants that damage kidneys are called as Hepatotoxic.
- In the light of the above statements, choose the most appropriate answer from the options given below :
- Both **Statement I** and **Statement II** are correct
 - Both **Statement I** and **Statement II** are incorrect
 - Statement I** is correct but **Statement II** is incorrect
 - Statement I** is incorrect but **Statement II** is correct

38. Which of the following are related to 'source reduction' in integrated solid waste management?

- A. Reduced Toxicity B. Composting
C. Less Packaging D. Landfill
E. More durable products

Choose the **correct** answer from the options given below :

- (1) (A), (B) and (C) Only (2) (A), (C) and (E) Only
(3) (B), (C) and (D) Only (4) (C), (D) and (E) Only

39. Depletion of stratospheric ozone results into

- A. Increase in the ultraviolet radiation
B. Increase in infra-red radiation
C. Increase in skin cancer
D. Significant impact on terrestrial flora and fauna
E. Decrease in cataract cases

Choose the **correct** answer from the options given below :

- (1) (A), (B), (C) and (D) Only
(2) (A), (C) and (D) Only
(3) (B), (C), (D) and (E) Only
(4) (A), (B) and (C) Only

40. Nitrous Oxide (N₂O).

- A. causes acid rain B. is laughing gas
C. is carcinogenic D. is a green house gas
E. is a criterion air pollutant

Choose the **correct** answer from the options given below :

- (1) (A), (D) and (E) Only (2) (C), (D) and (E) Only
(3) (B) and (D) Only (4) (A) and (E) Only

41. The Michael Sadler commission was constituted to study the problems of :

- (1) Bombay University
(2) Madras University
(3) Poona University
(4) Calcutta University

42. The National Education Policy, 1986 recommended the establishment of _____ in the country.

- (1) Community colleges
(2) Open universities
(3) National universities of eminence
(4) Discipline-wise universities

43. Given below are two statements:

Statement I: Curzon's university reforms reflected the official attitude towards the spread of higher education in India.

Statement II: There was no resistance to the Indian universities Bill piloted by Curzon from any section of Indians.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both **Statement I** and **Statement II** are true
(2) Both **Statement I** and **Statement II** are false
(3) **Statement I** is true but **Statement II** is false
(4) **Statement I** is false but **Statement II** is true

44. Match List-I with List-II.

List-I (Ancient thinkers)		List-II (Area of study)	
(A)	Panini	(I)	Astronomy
(B)	Jivaka	(II)	Grammar
(C)	Bhaskara	(III)	Medicine
(D)	Kanada	(IV)	Physics

Choose the **correct** answer from the options given below :

- (1) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)
(2) (A)-(II), (B)-(I), (C)-(III), (D)-(IV)
(3) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)
(4) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

45. Match List-I with List-II.

List-I (Name of University)		List-II (Year of Establishment)	
(A)	Bombay University	(I)	1916
(B)	Delhi University	(II)	1943
(C)	Utkal University	(III)	1922
(D)	S.N.D.T. Women University	(IV)	1857

Choose the **correct** answer from the options given below :

- (1) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)
(2) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)
(3) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)
(4) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

Read the passage and answer the questions that follow :

Science and the techniques to which it has given rise have changed human life during the last hundred and fifty years more than it had been changed since men took to agriculture, and the changes that are being wrought by science continue at an increasing speed. There is no sign of any new stability to be attained on some scientific plateau. On the contrary, there is every reason to think that the revolutionary possibilities of science extend immeasurably beyond what has so far been realised. Can the human race adjust itself quickly enough to these vertiginous transformations, or will it, as innumerable former species have done, perish from lack of adaptability? The dinosaurs were, in their day, the lords of creation, and if there had been philosophers among them, not one would have foreseen that the whole race might perish. But they became extinct because they could not adapt themselves to a world without swamps. In the case of man and science there is a wholly new factor, namely that man himself is creating the changes of environment to which he will have to adjust himself with unprecedented rapidity. But, although man through his scientific skill is the cause of the changes of environment, most of these changes are not willed by human beings. Although they come about through human agencies, they have, or at any rate have had so far, something of the inexorable inevitability of natural forces. Whether Nature dried up the swamps or men deliberately drained them, makes little difference as regards the ultimate result. Whether men will be able to survive the changes of environment that their own skill has brought about is an open question. If the answer is in the affirmative, it will be known some day; if not, not. If the answer is to be in the affirmative, men will have to apply scientific ways of thinking to themselves and their institutions.

46. Dinosaurs became extinct because of :
- (1) A massive earthquake (2) Incessant rainfall
(3) Extreme cold (4) Lack of adaptability
47. In the scientific age who is responsible for the changing environment?
- (1) Nature (2) Man
(3) Natural Disasters (4) Wars
48. The main focus of the passage is on :
- (1) Extinction of Species (2) Extinction of Dinosaurs
(3) Science and Human life (4) Destruction of Forests
49. "If the answer is in the affirmative " _____ refers to the question related to:
- (1) Changes in environment
(2) Human survival
(3) Human extinction
(4) Changes in scientific approach

50. According to the passage the following are true :
- A. Science is bringing about rapid changes.
B. There is every likelihood that a scientific plateau will be reached.
C. Man will definitely perish like the Dinosaurs.
D. There are chances that mankind may become extinct due to lack of adaptability.
E. Changes in the environment made through human agencies have so far shown almost the same inexorable inevitability as that of natural forces in the past.

Choose the **correct** answer from the options given below :

- (1) (A), (D) and (E) Only
(2) (A), (B) and (E) Only
(3) (A) and (E) Only
(4) (B), (C) and (E) Only

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1. (2) Number of male employees in college A = $3040 \times \frac{13}{19} = 2080$

In college B = $3360 \times \frac{4}{7} = 1920$

In college C = $2920 \times \frac{9}{20} = 1314$

In college D = $2760 \times \frac{10}{23} = 1200$

In college E = $2840 \times \frac{13}{20} = 1846$

Total number of male employees in all college in
= $2080 + 1920 + 1314 + 1200 + 1846 = 8360$

2. (1) Number of female employees in college A

= $3040 \times \frac{6}{19} = 960$

In college B = $3360 \times \frac{3}{7} = 1440$

In college C = $2920 \times \frac{11}{20} = 1606$

In college D = $2760 \times \frac{13}{23} = 1560$

In college E = $2840 \times \frac{7}{20} = 994$

Average number of female employees in all college
= $\frac{960 + 1606 + 1440 + 1560 + 994}{5} = \frac{6560}{5} = 1312$

3. (4) Number of male employees working in college A and C

= $3040 \times \frac{13}{19} + 2920 \times \frac{9}{20}$

= $2080 + 1314 = 3394$

Number of female employees working in college B and D

= $3360 \times \frac{3}{7} + 2760 \times \frac{13}{23}$

= $1440 + 1560 = 3000$

∴ Required increase % = $\frac{3394 - 3000}{3000} \times 100 = 13.13 \approx 13\%$

4 (2) Required ratio = $\frac{2760 \times \frac{13}{23}}{2840 \times \frac{7}{20}}$

= $\frac{1560}{994} = 780 : 497$

5 (2) Required increase% = $\frac{2920 - 2760}{2760} \times 100 = 5.79 \approx 6\%$

6. (2) Canva is a powerful design tool specifically designed for creating visual content such as infographics, posters, and presentations. It provides a wide range of templates and design elements that are perfect for illustrating concepts like supply and demand.

7 (1) Statement I is correct: Norm referencing is used to interpret a score of an individual by comparing it with those of other individuals.

Statement II is also correct: Criterion referencing is used to interpret a person's performance by comparing it to some specified behavioral domain.

8. (3) The elements of observational learning are (A), (B), (D) and (E) Only.

A. **Attention:** The learner must pay attention to the model's behavior.

B. **Retention:** The learner must be able to remember the behavior that was observed.

D. **Production:** The learner must be able reproducing the observed behavior

E. **Motivation:** The learner must have a reason or motivation to adopt the behavior.

9. (4) In the 'VARK' model of learning styles, 'A' stands for Aural, meaning a preference for learning through listening.

10. (2) When a very bright light is incident on our eyes, we immediately close our eyelids. This is an example of Reflex actions

Reflex actions are automatic and rapid responses to stimuli that do not involve conscious thought, such as closing our eyelids in response to bright light.

11. (2) The disadvantages of open-ended questions in survey research include:

B. They are time-consuming for interviewers to administer because the responses need to be recorded verbatim and then coded and analyzed.

C. They require greater effort from respondents because respondents must think about their answers and articulate them in their own words, which can be more demanding than selecting from predefined options.

12. (4) In APA style, the reference for a journal article is typically formatted as follows:

C. Name of the author

D. Title of the article

B. Volume number (and issue number, if applicable)

A. DOI number (if available)

13. (2) A research study is said to have heuristic value if it stimulates further investigation and research. Heuristic value refers to the ability of a study to inspire new questions, hypotheses, and subsequent studies.

14. (2) The number of marks scored in an examination is an example of a discrete variable because it involves countable values (e.g., 1, 2, 3, etc.). The marks are not measured on a continuous scale but rather as distinct, separate values.
15. (1) **Statement I:** Medical research often involves procedures that can cause pain or distress in animals, such as surgical interventions, drug testing, or disease modeling.
Statement II: Psychological research typically involves less invasive procedures, such as behavioural observations or cognitive tests, which usually result in minimal discomfort for the animals involved.
16. (4) A photograph is considered an "open text" because its interpretation is open to various meanings. Unlike closed texts, which have a fixed, singular meaning, open texts allow viewers to derive multiple interpretations based on their perspectives, experiences, and contexts. The visual elements in a photograph can evoke different emotions and thoughts, making it an open medium for interpretation.
17. (3) The Buddhist concept of communication is marked by "multi-causality." This aligns with the principle of dependent origination (Pratītyasamutpāda), which suggests that all events and phenomena arise from multiple causes and conditions. In this view, communication is not linear or singular but results from a complex interplay of various factors and influences.
18. (2) Cyber communication is considered a "palpable reality" at present. It is an integral part of everyday life, facilitating real-time interactions, social connections, and business transactions globally. Unlike fictitious or futuristic realities, cyber communication is tangible and widely experienced, making it a conventional and significant aspect of modern life.
19. (1) The primary objectives of public policy sphere communication include involving citizens in idea generation, policy critique, and innovative policy practices. These activities promote active public engagement, ensuring that diverse viewpoints and innovative solutions contribute to the policymaking process. Ignoring oppositional views and promoting stereotypes, as mentioned in options (D) and (E), are counterproductive to the goals of inclusive and democratic communication in the public policy sphere. Therefore, option (1) correctly identifies the objectives that align with effective public policy communication.
20. (1) Western communication models often focus on analyzing the relationship between communicative elements individually, considering each element's specific role and impact. On the other hand, the Buddhist concept of communication emphasizes the importance of context and the interdependence of communicative elements, viewing them as part of a holistic system. Both statements accurately describe these differing approaches to communication analysis, making both statements true.
21. (1) Man's one day work = $\frac{1}{6}$
 Woman's one day work = $\frac{1}{9}$
 Boy's one day work = $\frac{1}{18}$
 Let 'B' no. of boys needed

$$\Rightarrow \frac{3}{18} + \frac{2}{18} + B\left(\frac{1}{18}\right) = 1 \quad (\text{LCM of 6, 9, 18} = 18)$$

$$\Rightarrow \frac{B}{18} = \frac{13}{18} \therefore B = 13$$

Hence, to complete work in one day 13 boys are required.

22. (3) (i) B is the brother-in-law of C's mother.

Hence, B is a male.

(ii) C is the father of F.

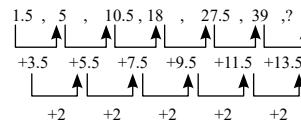
Hence, C is a male.

(iii) D is the father of A and F is the grandson of D.

Hence, D and F are male.

So, the remaining two members A and E are female.

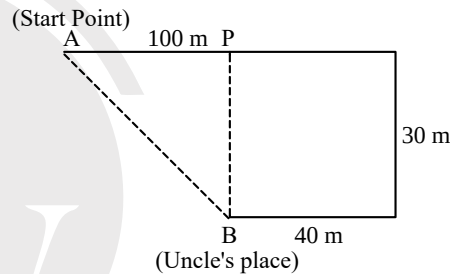
23. (2)



$$? = 39 + 13.5$$

$$? = 52.5$$

24. (1)



Shortest distance between starting point and uncle's place

$$= [(60)^2 + (30)^2]^{1/2}$$

$$= [3600 + 900]^{1/2}$$

$$= 30\sqrt{5} \text{ m}$$

25. (1) Logic:- Succeeding number's cube of the cube root of first number

$$\text{As, } 1000 \text{ i.e., } 10^3 \rightarrow (\text{Succeeding number's cube})$$

$$\rightarrow 11^3 = 1331$$

$$\text{and, } 125 \text{ i.e., } 5^3 \rightarrow 6^3 = 216$$

Similarly,

$$725 \text{ i.e., } 9^3 \rightarrow 10^3 = 1000$$

Hence, the correct answer is 1000.

26. (1) Statements B and D are contradictory to each other.

Statement B:- No plants are fungi.

This statements indicates that there is no relation between plant and fungi.

Statement D:- Some plants are fungi.

This statement states that there is at least one plant is a fungus.

So, the statement B and D both are not possible at a time.

27. (2) According to the given statement "All trucks are polluting vehicles" is declared false. This directly tells that not all trucks fit into the category of polluting vehicles.
Therefore, "Some trucks are not polluting vehicles" is correct.
28. (4) The original statement, "Some philosophers are not idealists," implies that within the set of philosophers, some individuals do not belong to the set of idealists. To find the equivalent statement, we need to reformulate it while maintaining the same logical structure.
Statement (1): "Some philosophers are not non-idealists" contradicts the original statement. If some philosophers are not non-idealists, it implies they are idealists, which is the opposite of the original meaning.
Statement (2): "No idealists are philosophers" implies that none of the idealists are philosophers, which is stronger than the original statement and not equivalent.
Statement (3): "Some non-idealists are not philosophers" introduces non-idealists and claims some of them are not philosophers. This changes the subject of the statement and is not equivalent.
Statement (4): "Some non-idealists are not non-philosophers" is equivalent to saying that some non-idealists are philosophers, which aligns with the original statement: "Some philosophers are not idealists."
29. (4) The **Fallacy of Division** occurs when one assumes that what is true for the group as a whole must also be true for each part. In this case, just because Indian pharmaceutical companies as a whole are profitable does not necessarily mean that every individual pharmaceutical company in India is profitable.
30. (4) **Statement I** is false because the Aristotelian syllogism is primarily deductive, not inductive, and focuses on formal logic.
Statement II is true because the **Nyāya syllogism** combines both **deductive** and **inductive** reasoning and includes both **formal** (structure) and **material** (content) aspects.
31. (2) An Intranet is a private network used within an organization to share company information and resources among employees. It enhances internal communication, collaboration, and efficiency by allowing employees to easily access and share important information. Unlike the Internet, which is accessible to the public, an Intranet is restricted to authorized personnel within the organization, ensuring that sensitive information is kept secure and is not shared with customers, competitors, or former employees.
32. (4) All three formulas (A), (B), and (C) correctly use the COUNTIFS function to count the number of product codes from UP that are of Grade "A" quality. The slight variations in the formatting and ordering of arguments in the formulas do not affect their correctness, as COUNTIFS can handle criteria in any order as long as they are correctly paired with their corresponding ranges. Therefore, all three options are correct.
33. (4) **Statement I is false.** Disk defragmentation is a process that reduces fragmentation by reorganizing scattered pieces of data so that files are stored contiguously, not fragmented.
Statement II is true. A device controller communicates with the operating system through a device driver, which is a software component that allows the OS to interact with the hardware device.
34. (2) **Statement (A):** This is correct because AI algorithms, such as machine learning models, are employed to detect and filter out spam emails by analyzing patterns and content in email messages.
Statement (B): AI systems are capable of analyzing uncertain information. This is correct as AI systems use techniques such as fuzzy logic, probabilistic models, and machine learning to handle and analyze data that may be incomplete or uncertain.
Statement (C): This is incorrect. The description given in Statement (C) refers to a Geographic Information System (GIS), which is used for geospatial data management, not specifically to AI. AI can be integrated with GIS for enhanced analysis, but AI itself is not defined by this function.
35. (4) **Statement (A):** Spell checker and Thesaurus are available features in MS-WORD. The spell checker helps in correcting spelling errors, while the thesaurus provides synonyms and antonyms for words.
Statement (B): Page Orientation is an available feature in MS-WORD. It allows users to set the page layout to either portrait or landscape orientation.
Statement (C): Mail Merge is an available feature in MS-WORD. It is used to create multiple documents, such as letters or labels, by combining a standard template with a data source.
Statement (D): Slide Transitions are not a feature of MS-WORD. Slide transitions are specific to presentation software like MS PowerPoint, where they are used to create visual effects when moving from one slide to another.
36. (4) Tuberculosis is a bacterial infection caused by Mycobacterium tuberculosis, primarily affecting the lungs and is spread through airborne particles, not through polluted water.
Diseases caused by polluted water include:
Cholera: Caused by Vibrio cholerae bacteria, spread through contaminated water.
Dysentery: Caused by bacteria or amoeba in contaminated water.
Typhoid: Caused by Salmonella typhi bacteria, spread through contaminated water and food.
37. (3) **Statement I:** Toxic chemicals do often injure human organs and organ systems. This statement is correct.
Statement II: Toxicants that damage kidneys are called nephrotoxic, not hepatotoxic. "Hepatotoxic" refers to toxicants that damage the liver. This statement is incorrect.
38. (2) Source reduction in integrated solid waste management refers to practices that reduce the amount and toxicity of waste generated. This can include:
A. Reduced Toxicity: Decreasing the hazardous nature of waste produced.
C. Less Packaging: Minimizing packaging materials to reduce waste.
E. More durable products: Designing and using products that last longer and thus generate less waste over time.
Composting (B) and landfilling (D) are methods of managing waste after it has been generated, not practices that reduce the amount or toxicity of waste at the source.
39. (2) Depletion of stratospheric ozone results in the following consequences:

- A. Increase in the ultraviolet radiation:** The ozone layer absorbs and blocks most of the sun's harmful ultraviolet (UV) radiation. Depletion of the ozone layer leads to an increase in the amount of UV radiation reaching the Earth's surface.
- C. Increase in skin cancer:** Increased exposure to UV radiation is a major risk factor for skin cancer.
- D. Significant impact on terrestrial flora and fauna:** Higher levels of UV radiation can harm plants and animals, affecting their health and ecosystems.

Infra-red radiation (B) is not directly related to the depletion of the ozone layer, and a decrease in cataract cases (E) is incorrect because increased UV radiation actually leads to more cataract cases, not fewer.

40. (3) Nitrous Oxide (N_2O):

(B) is laughing gas: Nitrous oxide is commonly known as laughing gas and is used as an anesthetic in dentistry and surgery.

(D) is a greenhouse gas: Nitrous oxide is a potent greenhouse gas with significant global warming potential.

It is not directly responsible for causing acid rain (A), it is not classified as carcinogenic (C), and it is not one of the six criteria air pollutants identified by the EPA (E).

41. (4) The Michael Sadler Commission, also known as the Calcutta University Commission, was constituted in 1917 to study and address the problems and make recommendations for the improvement of Calcutta University.
42. (2) The National Education Policy (NEP) of 1986 recommended the establishment of open universities in the country to promote distance education and provide opportunities for higher education to a larger population, particularly those who are unable to attend traditional universities.
43. (3) **Statement I:** Curzon's university reforms did reflect the official attitude towards the spread of higher education in India. His reforms aimed to improve the quality of higher education and ensure better control over universities.

Statement II: There was significant resistance to the Indian Universities Bill piloted by Curzon from various sections of Indians, including students, teachers, and nationalist leaders who saw the reforms as an attempt to control and limit the spread of higher education.

44. (3) **(A) Panini - (II) Grammar:** Panini was an ancient Indian scholar who is most famous for his work on Sanskrit grammar, particularly his treatise called *Ashtadhyayi*.

(B) Jivaka - (III) Medicine: Jivaka was a renowned physician in ancient India, often associated with traditional medicine and the healing practices during the time of Buddha.

(C) Bhaskara - (I) Astronomy: Bhaskara I was an ancient Indian mathematician and astronomer known for his contributions to the understanding of planetary movements and celestial bodies.

(D) Kanada - (IV) Physics: Kanada was an ancient Indian philosopher who founded the Vaisheshika school of philosophy and is credited with pioneering ideas related to atomism and physics.

45. (1) **(A) Bombay University - (IV) 1857:** The University of Mumbai, formerly known as Bombay University, was established in 1857, making it one of the oldest universities in India.

(B) Delhi University - (III) 1922: The University of Delhi, one of the premier universities in India, was established in 1922.

(C) Utkal University - (II) 1943: Utkal University, located in Odisha, was established in 1943 and is one of the oldest universities in the state.

(D) S.N.D.T. Women University - (I) 1916: The Shreemati Nathibai Damodar Thackersey (S.N.D.T.) Women's University, established in 1916, is the first women's university in India.

46. (4) The passage mentions that dinosaurs became extinct because they could not adapt themselves to a world without swamps, indicating a lack of adaptability.
47. (2) The passage states that man himself is creating the changes in the environment to which he will have to adjust, implying that in the scientific age, man is responsible for the changing environment.
48. (3) The passage discusses how science has changed human life and the environment, and the need for humans to adapt to these changes.
49. (2) The passage is questioning whether humans will be able to survive the changes in the environment that their own skill has brought about, which relates to human survival.
50. (1) The passage states that science is bringing about rapid changes (A), and there are chances that mankind may become extinct due to lack of adaptability (D). It also mentions that changes in the environment made through human agencies have shown almost the same inevitability as natural forces in the past (E). However, it does not suggest that a scientific plateau will be reached (B) or that man will definitely perish like the dinosaurs (C).

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